Problem Set 18

Areas Between Curves I

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Problem 2

(b)

$$y = \sqrt{x+3}, y = \frac{x+3}{2}$$
$$\sqrt{x+3} = \frac{x+3}{2} \implies (x-1)(x+3) = 0$$
$$A = \int_{-3}^{1} \sqrt{x+3} - \frac{x+3}{2}\,dx$$
$$= \int_{0}^{4} \sqrt{u} - \frac{u}{2}\,du$$
$$= \frac{2}{3}u^{\frac{3}{2}} - \frac{x^2}{4}\Big|_0^4 = \boxed{\frac{4}{3}}$$

(d)

$$y = \sec^2 x, y = 8\cos x$$
$$\sec x = 2 \implies x = \pm\frac{\pi}{3}$$
$$A = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} 8\cos x - \sec^2 x\,dx$$
$$= 8\sin x - \tan x\Big|_{-\frac{\pi}{3}}^{\frac{\pi}{3}}$$
$$= 8\sqrt{3} - 2\sqrt{3} = \boxed{6\sqrt{3}}$$

(f)

$$x = 2y^2, x = 4 + y^2$$
$$y = 2 \implies x = 8$$
$$A = \int_{-2}^{2} 4 + y^2 - 2y^2\,dy$$
$$= \int_{-2}^{2} 4 - y^2\,dy$$
$$= 4y - \frac{y^3}{3}\Big|_{-2}^{2} = \boxed{\frac{32}{3}}$$

(h)

$$y = \cos x, y = 1 - \frac{2x}{\pi}$$
$$x = 0 \text{ or } \frac{\pi}{2} \text{ or } \pi$$
$$A = 2\int_0^{\frac{\pi}{2}} \frac{2x}{\pi} - 1 + \cos x \, dx$$
$$= 2(\sin x - x + \frac{x^2}{2})\Big|_0^{\frac{\pi}{2}} = \boxed{2 - \frac{\pi}{2}}$$

Problem 3

(b)

$$\sin x - \cos 2x = 0$$
$$\sin x + 2\sin^2 x - 1 = 0$$
$$(2\sin x - 1)(\sin x + 1) = 0$$
$$\sin x = \frac{1}{2} \implies x = \frac{\pi}{6}$$
$$\int_0^{\frac{\pi}{2}} |\sin x - \cos 2x| \, dx$$
$$= \int_0^{\frac{\pi}{6}} \cos 2x - \sin x \, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin x - \cos 2x \, dx$$
$$= \frac{1}{2}\sin 2x + \cos x\Big|_0^{\frac{\pi}{6}} + \frac{1}{2}\sin 2x + \cos x\Big|_{\frac{\pi}{2}}^{\frac{\pi}{6}}$$
$$= \frac{\sqrt{3}}{2} + \sqrt{3} - 1 - 0 = \boxed{\frac{3\sqrt{3}}{2} - 1}$$

Problem 4

(a)

$$\sqrt{x} = x - 2 \implies x = 4$$
$$A = \int_0^2 \sqrt{x} \, dx + \int_2^4 \sqrt{x} - x + 2 \, dx$$
$$= \frac{2}{3}x^{\frac{3}{2}}\Big|_0^2 + \frac{2}{3}x^{\frac{3}{2}} - \frac{x^2}{2} + 2x\Big|_2^4$$
$$= \frac{16}{3} - 8 + 8 + 2 - 4 = \boxed{\frac{10}{3}}$$

(b)

$$A = \int_0^2 y + 2 - y^2 \, dy$$

$$= \frac{y^2}{2} + 2y - \frac{y^3}{3}\Big|_0^2$$
$$= 2 + 4 - \frac{8}{3} = \boxed{\frac{10}{3}}$$

(c) The second method is simpler. The general rule is to choose integration where fewer pieces are involved.

Problem 5

$$A = \int_{-1}^{0} -\frac{5}{3}x + \frac{13}{3} + 6x\,dx + \int_{0}^{2} -\frac{5}{3}x + \frac{13}{3} - \frac{1}{2}x\,dx$$
$$= \frac{13}{6}x^2 + \frac{13}{3}x\Big|_{-1}^{0} - \frac{13}{12}x^2 + \frac{13}{3}x\Big|_0^2$$
$$= \frac{13}{3} - \frac{13}{6} - \frac{13}{3} + \frac{26}{3} = \boxed{\frac{13}{2}}$$
$$A = \frac{1}{2}|2 \cdot 6 - (-1) \cdot 1| = \boxed{\frac{13}{2}}$$

Problem 6

$$A = \int_0^1 \cos^2(\frac{\pi x}{4}) - \sin^2(\frac{\pi x}{4})\,dx$$
$$= \int_0^1 \cos\frac{\pi x}{2}\,dx$$
$$= \frac{2}{\pi}\sin\frac{\pi x}{2}\Big|_0^1 = \boxed{\frac{2}{\pi}}$$

Problem 7

(a) $L(x) = x$ represents perfect equality.

$$G = 2\int_0^1 (x - L(x))\,dx \geq 0$$
$$G = 1 - 2\int_0^1 L(x)\,dx \implies G < 1$$
$$\boxed{0 \leq G < 1}$$

(b)

$$G = 2\int_0^1 (x - x^3)\,dx = 2(\frac{x^2}{2} - \frac{x^4}{4})\Big|_0^1 = \boxed{\frac{1}{2}}$$
$$G = 2\int_0^1 (x - \frac{e^x - 1}{e - 1})\,dx = 2(\frac{x^2}{2} - \frac{e^x - x}{e - 1})\Big|_0^1 = \boxed{\frac{3 - e}{e - 1}}$$

The second function of $L(x)$ describes more equality.

(c)

$$G = 2\int_0^1 (x - x^p)\,dx = 2(\frac{x^2}{2} - \frac{x^{p+1}}{p+1})\Big|_0^1 = \boxed{\frac{p-1}{p+1}}$$

$$\lim_{p\to\infty} G = \lim_{p\to\infty} \frac{p-1}{p+1} = \boxed{1}$$

It describes a society with absolute inequality.

Problem 8

(a)

$$y = (a+b)(x-a) + a^2 = (a+b)x - ab$$

$$A = \int_a^b (a+b)x - ab - x^2\,dx$$

$$= \left(\frac{a+b}{2}x^2 - abx - \frac{1}{3}x^3\right)\Big|_a^b$$

$$= \frac{a+b}{2}(b^2 - a^2) + ab(a-b) + \frac{1}{3}(a^3 - b^3)$$

$$= \boxed{\frac{(b-a)^3}{6}}$$

(b)

$$y = x^2 \implies y' = 2x$$

$$a + b = 2x \implies x = \frac{a+b}{2}$$

$$A = \frac{1}{2}(b-a)\left(\frac{(a+b)^2}{2} - ab - \frac{(a+b)^2}{4}\right)$$

$$= \boxed{\frac{(b-a)^3}{8}}$$

(c)

$$y = 2a(x-a) + a^2 = 2ax - a^2$$

$$y = 2b(x-b) + b^2 = 2bx - b^2$$

$$b^2 - a^2 = 2(b-a)x \implies x = \frac{a+b}{2}$$

$$A = \int_a^{\frac{a+b}{2}} (x-a)^2\,dx + \int_{\frac{a+b}{2}}^b (x-b)^2, dx$$

$$= \frac{(x-a)^3}{3}\Big|_a^{\frac{a+b}{2}} + \frac{(x-b)^3}{3}\Big|_{\frac{a+b}{2}}^b$$

$$= \boxed{\frac{(b-a)^3}{12}}$$

(d)

$$y = x^3 \implies y' = 3x^2$$
$$y = 3t^2(x - t) + t^3 = 3t^2x - 2t^3$$
$$x^3 - 3t^2x + 2t^3 = 0$$
$$(x - t)^2(x + 2t) = 0 \implies \boxed{x = -2t}$$
$$A = \int_{-2t}^{t} x^3 - 3t^2x + 2t^3 \, dt$$
$$= \frac{x^4}{4} - \frac{3t^2x^2}{2} + 2t^3x \Big|_{-2t}^{t}$$
$$= \boxed{\frac{27}{4}t^4}$$